Heidelberg University
Institute for Theoretical Physics
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Tristan Bereau

Employment

Heidelberg University, *Institute for Theoretical Physics*
Professor (W3) *2023–*

IMC Trading B.V.
Quant *2021–2023*

University of Amsterdam, *Van 't Hoff Institute for Molecular Sciences, Informatics Institute*
Assistant Professor *2020–2021*

Max Planck Institute for Polymer Research, *Theory Group*
Emmy Noether group leader *2016–2019*
Group leader *2014–2016*

University of Basel and Novartis
Postdoc *2012–2013*

Education

Carnegie Mellon University, *Department of Physics*
Ph.D. *2011*

École Polytechnique Fédérale de Lausanne
BSc. Physics *2006*

Secondary appointments

Spokesperson, Artifical Intelligence (AI) Board, Heidelberg University. *2024–*
Member, IWR Heidelberg. *2023–*
Member, SIMPLAIX. *2023–*
Editorial Board member, Computational Science and Engineering, Springer Nature. *2023–*
Committee member, Diversity & Inclusion, HIMS, UvA. *2021*
Committee member, National Agenda for Computational Sciences, NWO. *2021*
Data-driven discovery and design
HIMS PhD Lectures, UvA. *2020–2021*
Guest Editor, APL Materials Special Issue. *2020*
"Discovering Patterns in Disorder: Machine Learning for Fluctuating Mesoscopic Materials," together with Dr. Alpha Lee and Prof. Daan Frenkel (U. Cambridge)
Editorial Board member, Machine Learning: Science and Technology, IOP Publishing. *2019–*

Steering-committee member, Carl-Zeiss-Stiftung on Algorithmic Intelligence. *2019–2020*

Board member, TRR146. *2018–2020*

Deputy member representative, FAIR-DI e.V. (FAIR Data Infrastructure). *2018–*
Pillar C: Soft-matter and biomolecular simulations

Deputy member representative, Max Planck Network BiGmax. *2017–*
Project Area 4: Learning thermodynamic properties of materials

Co-Head, TRR146 Integrated research training group (IRTG). *2017–2020*

Junior PI, Max Planck Graduate Center (graduate school MPIP/JGU Mainz). *2017–2020*

Awards

"Hochschullehrer-Nachwuchs-Workshop 2019," Marl, Germany. *2019*
Invitation by GDCh Makromolekulare Chemie to selected young university lecturers

"Machine Learning for Physics and the Physics of Learning," CA USA. *09–12/2019*
Invited core participant of IPAM Long Program

Examination authorization (Prüfungsberechtigung), Physics, JGU Mainz. *2018*

Emmy Noether Program, Deutsche Forschungsgemeinschaft (DFG). *2016*
Independent research group, Equivalence to Habilitation

Astrid and Bruce McWilliams Fellowship, CMU, Pittsburgh, PA USA. *2010*
1-year full tuition and stipend scholarship

Student Travel Award – Biophysical Society Meeting, San Francisco, CA. *2010*

Guy C. Berry Graduate Research Award – Mellon College of Science, CMU. *2009*

Funding

PhD project, Deutsche Forschungsgemeinschaft (DFG). *2025–2028*
Generative machine learning in the grand canonical ensemble (€0.2M)

Field of Focus 2 of Heidelberg University. *2024*

FAIRmat: consortium of the German Research-Data Infrastructure NFDI. *2022–*
Area C2: Multiscale modeling

Roche Postdoctoral Fellow: Machine Learning for Crystal Structure Prediction. *2021–2022*
With F. Hoffmann-La Roche AG, Avant-garde Materials Simulation GmbH
Role: Co-supervisor

Simulating catalysis: Multiscale embedding of machine learning potentials. *2021–2022*
EPSRC (UK) grant
Role: Project partner

Carl-Zeiss-Stiftung "Algorithmic Intelligence as Emergent Phenomenon". *2019–2023*
One of 4 core co-PIs (Consortium: €3.0M)
Statistical predictability of physical systems (€0.2M)

Max Planck Network BiGmax on "Big Data in Materials Science". *2018–2022*
Automated analysis of atom force probe detector maps (€0.2M)
Learning thermodynamic properties of materials (€0.2M)

Emmy Noether Program, Deutsche Forschungsgemeinschaft (DFG). *2016–2020*
Independent research group (€1.3M)
Topic: Importance Sampling in Chemical Space

TRR146 "Multiscale Modeling in Soft Matter". *2014–2022*
A6: Dynamics in multiscale simulations (€0.2M)
B7: Machine learning in multiscale simulations (€0.2M)

Academic interviews

Theoretical Physics in soft condensed matter. University of Heidelberg. *05/2022*
W3 professor position. Offer, accepted

Machine learning in the physical sciences. University of Stuttgart. *06/2021*
W3 professor position

Department of Physics. University of Stuttgart. *11/2020*
W3 professor position

Computational Soft Matter. University of Amsterdam. *03/2019*
Tenure-track assistant professor position. Offer, accepted

Theory of Condensed Matter Physics. University of Konstanz. *11/2018*
Tenure-track professor position

Data science in the natural sciences. SISSA, Trieste. *02/2018*
Tenure-track professor position

Computational materials modeling. SEAS, Harvard University. *04/2017*
Tenure-track assistant professor position

Teaching (courses)

MVBP2 Theoretical Biophysics. Heidelberg University. *Summer 2025*

MVSpec Soft matter physics. Heidelberg University. *Summer 2025*
Co-taught with Dr. Falko Ziebert

MKTP1 Theoretical statistical physics. Heidelberg University. *Winter 2024*

MVComp2 Computational statistics and data analysis. Heidelberg University. *Summer 2024*
Co-taught with Prof. Daniel Durstewitz

MVSpec Soft matter physics. Heidelberg University. *Summer 2024*
Co-taught with Dr. Falko Ziebert

MVSpec Machine learning in the biomolecular world. Heidelberg University. *Summer 2024*
Co-taught with Prof. Rebecca Wade

MVComp2 Computational statistics and data analysis. Heidelberg University. *Winter 2023*

Generative learning for molecular systems. Heidelberg University. *Winter 2023*

Introduction Computational Science. University of Amsterdam. *Spring 2021*
Co-taught with Dr. Valeria Krzhizhanovskaya

Statistical Thermodynamics of Complex Molecular Systems. University of Amsterdam. *Fall 2020*
Co-taught with Prof. Peter Bolhuis, Dr. Bernd Ensing

Advanced computer simulation techniques. JGU Mainz. *Spring 2019*
Co-taught with Dr. Giovanni Settanni, Dr. Omar Valsson

Computer simulation techniques. JGU Mainz. *Fall 2018*
Co-taught with Dr. Giovanni Settanni, Dr. Peter Virnau

Polymer physics and soft-matter theory. JGU Mainz. *Fall 2018*
Co-taught with Prof. Burkhard Dünweg, Dr. Sara Jabbari-Farouji

Advanced computer simulation techniques. JGU Mainz. *Spring 2018*
Co-taught with Dr. Denis Andrienko, Dr. Giovanni Settanni

Polymer physics and soft-matter theory. JGU Mainz. *Fall 2017*
Co-taught with Prof. Burkhard Dünweg, Dr. Sara Jabbari-Farouji

Polymer physics and soft-matter theory. JGU Mainz. *Fall 2016*
Co-taught with Prof. Burkhard Dünweg, Dr. Sara Jabbari-Farouji

Advanced computer simulation techniques. JGU Mainz. *Spring 2016*
Co-taught with Dr. Denis Andrienko, Dr. Giovanni Settanni

Electronic structure calculations. University of Basel. *Fall 2012*
Co-taught with Prof. Markus Meuwly

Teaching (guest lectures)

Machine learning for physics and astronomy. UvA/Vu (Amsterdam). *10/2020*
Machine learning for coarse-grained simulations

Biomolecular simulations. UvA (Amsterdam). *05/2020*
Machine learning for coarse-grained simulations

Teaching (summer schools)

MolSim 2020. UvA (Amsterdam). *01/2020*

CECAM: "Applied mathematics and machine learning perspectives on Big Data". Mainz. *10/2019*

CECAM/TRR School on Machine Learning. Mainz. *09/2018*

CCCS School on machine learning in atomistic simulations. Vallico Sotto, Italy. *05/2017*

CCCS School on coarse-graining. Engelberg, Switzerland. *03/2017*

ESPResSo++/VOTCA tutorial. Schloss Waldthausen, Mainz. *10/2014*

Tutoring

CECAM/TRR School on Machine Learning. Mainz. *09/2018*

CCCS School on coarse-graining. Engelberg, Switzerland. *03/2017*

ESPResSo++/VOTCA tutorial. Schloss Waldthausen, Mainz. *10/2014*

ESPResSo workshop. University of Stuttgart. *10/2010*

Workshop "Coarse-Grained Biomolecular Modeling". Levi, Finland. *03/2010*

Workshop organization

2025 ESPResSo Summer School. Stuttgart, Germany. *10/2025*

3rd SIMPLAIX Workshop on Machine Learning for Multiscale Molecular Modeling. Heidelberg, Germany. *05/2025*

CCSC2024: Chemical Compound Space Conference 2024. Heidelberg Conference Center. *05/2024*

CECAM: "(Machine) learning how to coarse-grain". Online event (Zoom & Discord). *09/2020*
260 registered participants

CECAM: "Applied mathematics and machine learning perspectives on Big Data Problems in Computational Sciences,". CECAM. *10/2019*

Mainz Materials Simulation Days 2019. Max Planck Institute for Polymer Research. *06/2019*

International workshop for MaxNet on Big Data in Materials Science. Dresden. *04/2019*

CECAM/TRR School on Machine Learning. Mainz. *09/2018*
60 participants

CECAM: "New frontiers in particle-based multiscale modeling". CECAM. *09/2018*

Mainz Materials Simulation Days 2015. Max Planck Institute for Polymer Research. *06/2015*
100+ participants

ESPResSo++/VOTCA tutorial. Schloss Waldthausen. *10/2014*

Invited talks

Chalmers AI4Science Seminar. Gothenburg, Sweden. *02/2025*
online recording

TCP Workshop. Luxembourg, Luxembourg. *12/2024*

IWR Colloquium. Heidelberg, Germany. *11/2024*

Workshop: Physics of Intelligence and Machine Learning 2024. Heidelberg, Germany. *10/2024*

Workshop: Machine-Learning-Based Sampling in Lattice Field Theory and Quantum Chemistry. Bonn, Germany. *10/2024*

CECAM Flagship School: Simulating soft matter across scales. Stuttgart, Germany. *10/2024*

Multiscale Simulations of Soft Matter IV. Darmstadt, Germany. *09/2024*

CECAM Workshop: Bridging the Atomic-Mesoscale Gap for Complex Interfaces. Montpellier, France. *09/2024*

CECAM Workshop: Leveraging Machine Learning for Sampling Rare Events in Biomolecular Systems. Mainz, Germany. *09/2024*

31st CMD-General Conference of the Condensed Matter Division of the European Physical Society. Braga, Portugal. *08/2024*

IDA-BIFOLD-ML Group retreat. Berlin, Germany. *08/2024*

Physics department, Heidelberg University. Heidelberg, Germany. *06/2024*

Institute of Physical Chemistry, Karlsruhe Institute of Technology. Karslruhe, Germany. *05/2024*

ECT* Workshop Bridging Scales". Trento, Italy. *04/2024*

Institute for Computational Physics, University of Stuttgart. Stuttgart, Germany. *04/2024*

DWI - Leibniz Institute for Interactive Materials. Aachen, Germany. *03/2024*

CCRC1114 Colloquium. Berlin, Germany. *02/2024*

Institute of Physics, Freiburg University. Freiburg, Germany. *02/2024*

CECAM workshop: FAIR and TRUE Data in Soft Matter Simulations. Mainz, Germany. *09/2023*

Comp Chem Exchange, Merck KGaA. Darmstadt, Germany. *09/2023*
ELLIS unit Heidelberg. Heidelberg, Germany. *07/2023*
TMCQ2023. Seoul, South Korea. *06/2023*
MMSD 2023: Biology in soft matter. Mainz, Germany. *06/2023*
BiGmax School on Artificial Intelligence for Materials Science. Cap Roig, Spain. *05/2023*
SIMPLAIX Workshop "ML for Multiscale Molecular Modeling". Heidelberg, Germany. *05/2023*
CECAM Modeling Materials at Realistic time Scales. Berlin, Germany. *07/2022*
MMSML Workshop. Barcelona, Spain. *07/2022*
Lorentz "Accelerating the Understanding of Rare Events". Leiden, Netherlands. *09/2021*
CECAM "Non-covalent interactions in large molecules". Lausanne, Switzerland. *08/2021*
PASC21 Conference. Geneva, Switzerland. *07/2021*
CECAM "Local structure meets machine learning in soft matter systems". Switzerland. *06/2021*
Dutch Soft Matter meeting. virtual. *05/2021*
SIAM Conference "Mathematical Aspects of Materials Science". virtual. *05/2021*
GDCh Bunsen-Tagung 2021. virtual. *05/2021*
Plenary speaker
Department of Physics, University of Delaware. Newark, USA (virtual). *03/2021*
Minisymposium of the SFB CRC 1114. Berlin, Germany (virtual). *03/2021*
Minisymposium of the SFB TRR 102. Halle, Germany (virtual). *02/2021*
Institute for Materials Science, TU Dresden. Dresden, Germany (virtual). *12/2020*
Department of Chemistry, Boston University. Boston, USA (virtual). *10/2020*
CECAM Multiscale simulations of soft matter. virtual. *09/2020*
AI4Science Colloquium Series. University of Amsterdam, Amsterdam (virtual). *06/2020*
CECAM "Open Databases Integration for Materials Design". virtual. *06/2020*
Conference on A FAIR Data Infrastructure for Materials Genomics. virtual. *06/2020*
online recording
CompBioMed "Machine learning meets modelling and simulation methods". virtual. *03/2020*
online recording
PCoMS Seminar Series. Tohoku University, Sendai, Japan. *02/2020*
Department of Chemical Engineering, Kyoto University. Kyoto, Japan. *02/2020*
CANES Seminar Series. King's College London, London, United Kingdom. *02/2020*
Physics@Veldhoven. Veldhoven, Netherlands. *01/2020*
University of Marburg. Marburg, Germany. *11/2019*
E-CAM "Building a molecular foundry". virtual. *11/2019*
Leibniz Institute for Interactive Materials. Aachen, Germany. *11/2019*
IPAM "Machine Learning for Physics and the Physics of Learning". UCLA, USA. *10/2019*
online recording

CECAM "Beyond machine learning for quantum chemistry". Bremen, Germany. *10/2019*

"Hochschullehrer-Nachwuchs-Workshop 2019". Marl, Germany. *09/2019*

IPAM "Machine Learning for Physics and the Physics of Learning". UCLA, USA. *09/2019*

online recording

American Chemical Society (ACS) Fall Meeting 2019. San Diego, CA USA. *08/2019*

XXXI IUPAP Conference on Computational Physics (CCP2019). Hong Kong. *07/2019*

Advances in methods for multi-scale modelling,. Leiden, Netherlands. *06/2019*

"MolKin2019 Sampling, Design and Machine Learning". Berlin, Germany. *06/2019*

IPAM "Many-Particle Systems with Machine Learning". Lake Arrowhead, USA. *06/2019*

German Physical Society (DPG) meeting. Regensburg. *03/2019*

American Physical Society (APS) meeting. Boston, USA. *03/2019*

Machine Learning and Reverse engineering for Soft Materials. Leiden, Netherlands. *12/2018*

Integrating Molecular Simulation with Machine Learning. Leiden, Netherlands. *10/2018*

Max Planck Institute for Iron Research. Düsseldorf, Germany. *08/2018*

Modern Approaches to Coupling Scales In Materials Simulations. Lenggries, Germany. *07/2018*

Many-Particle Systems with Machine Learning. Lake Arrowhead, USA. *06/2018*

Computational Chemistry Days. Helsinki, Finland. *05/2018*

Keynote lecture

Theoretical Computational Chemistry Workshop. Engelberg, Switzerland. *05/2018*

SimTech Conference. Stuttgart, Germany. *03/2018*

German Physical Society (DPG) meeting. Berlin, Germany. *03/2018*

Cross-sectional symposium

Department of Applied Mathematics. TU Eindhoven, Eindhoven, The Netherlands. *02/2018*

Department of Chemistry. Ruhr University of Bochum, Bochum, Germany. *01/2018*

German Chemical Society (GDCh) Satellite Meeting. Berlin, Germany. *09/2017*

Symposium on Theoretical Chemistry (STC). University of Basel. *08/2017*

Department of Physical Chemistry. University of Münster. *07/2017*

Department of Physical Chemistry. University of Göttingen. *06/2017*

IPAM "Many-Particle Systems with Machine Learning". UCLA, USA. *11/2016*

Department of Chemistry. Free University of Brussels. *04/2016*

Department of Chemistry. University of Konstanz. *04/2016*

"Transferability Issues in Multiscale Modelings". Mainz. *12/2015*

Department of Chemistry. Freie Universität Berlin. *11/2015*

CCCS Symposium on Machine Learning. Basel, Switzerland. *11/2015*

Statistical Physics and Low Dimensional Systems. Pont-à-Mousson, France. *05/2015*

"Modeling Many-Body Interactions 2015". Lake Garda, Italy. *05/2015*

CSP Workshop. University of Georgia, Athens GA USA. *03/2015*

	Faculty of Natural Sciences. University of Groningen.	*06/2015*
	Mainz Materials Simulations Day (MMSD 2013). Mainz, Germany.	*06/2013*
	Department of Chemistry. Penn State.	*05/2013*
	Laboratoire de Biochimie Théorique. CNRS, Paris VII, France.	*01/2013*
	Forschungszentrum Jülich. Jülich, Germany.	*11/2012*
	Max Planck Institute for Polymer Research. Mainz.	*07/2010*
	INSERM. Paris, France.	*06/2010*
	Max Planck Institute for Polymer Research. Mainz.	*07/2009*
	Forschungszentrum Jülich. Jülich.	*07/2009*
Supervision	Ferdinand Hörstel. Master student (Chemistry).	*2025–2026*
	Gerrit Gerhartz. PhD student.	*2025–*
	Sander Hummerich. PhD student. co-supervised with Prof. Ullrich Köthe, IWR Heidelberg	*2025–*
	Daniel Hofmann. Master student.	*2025–*
	Magnus Schümann. Bachelor student.	*2025*
	Selina Pohl. Master student.	*2024–*
	Luis Vazquez-Salazar. Postdoc. SNSF postdoc fellowship	*2024–*
	Aleksander Durumeric. Postdoc. FI EMS interdisciplinary postdoc fellowship	*2024–2025*
	Leticia Gómez. Postdoc. co-supervised with Dr. Christopher Synatschke, MPIP Mainz	*2024–2025*
	Dominic Holst. Master student.	*2024–2026*
	Bálint Maté. Visiting Ph.D. student. SNSF fellowship	*2024*
	Daniel Nagel. Postdoc.	*2023–2026*
	Luis Walter. Ph.D. student.	*2023–*
	Osman Oezdemir. Master student.	*2023–2024*
	Diego van der Mast. Bachelor student.	*2021*
	Menno Bruin. Bachelor student.	*2020*
	Bernadette Mohr. Ph.D. student.	*2019–2023*
	Atreyee Banerjee. Postdoc.	*2019–2022*
	Yasemin Bozkurt Varolgüneş. Exchange Ph.D. student.	*2018–2019*
	Martin Girard. Postdoc. **Humboldt fellowship**	*2018–2021*

René Scheid. Master student. *2018–2019*

Timon Wittenstein. Bachelor student. *2018–2019*

Clemens Rauer. Postdoc. *2018–2019*

Arghya Dutta. Postdoc. *2018–2021*

Marc Stieffenhofer. Ph.D. student. *2018–2022*

MPGC fellowship

Christian Hoffmann. Diplom student. *2018–2019*

Bernadette Mohr. Diplom student. *2018–2019*

Alessia Centi. Postdoc. *2017–2019*

Kiran H. Kanekal. Ph.D. student. *2016–2020*

MPGC fellowship. Graduated with *Summa Cum Laude* (highest distinction)

Roberto Menichetti. Postdoc. *2016–2018*

Svenja Wörner. Ph.D. student. *2016–2020*

Marius Bause. Ph.D. student. *2016–2020*

MAINZ fellowship

Joseph F. Rudzinski. Postdoc. *2015–2019*

Humboldt fellowship

Chan Liu. Ph.D. student. *2014–2019*

Doctoral defense committees

Juan Viguera Diaz. Chalmers University of Technology (Opponent). *02/2026*

Morteza Shokrani. Heidelberg University. *02/2026*

Leon Lettermann. Heidelberg University. *01/2026*

Patrick Friebel. Heidelberg University. *11/2025*

Andrey Butkevich. Heidelberg University. *10/2025*

Nikhilesh Kottapalli. Heidelberg University. *07/2025*

Heiko Mager. Heidelberg University. *07/2025*

Weiyu Zhang. Heidelberg University. *04/2025*

Haoxiang Zhao. TU Eindhoven. *03/2025*

Lennart Rover. Heidelberg University. *01/2025*

Roman Remme. Heidelberg University. *01/2025*

Simon L. Krug. Technische Universität Berlin. *12/2024*

Oliver Drozdowski. Heidelberg University. *11/2024*

Gianmarco Camazzola. Heidelberg University. *10/2024*

Manuel Brenner. Heidelberg University. *07/2024*

Dominik Lemm. University of Vienna. *03/2024*

Robert Chojowski. Heidelberg University.	*01/2024*
Bernadette Mohr. University of Amsterdam.	*12/2023*
Onur Caylak. TU Eindhoven.	*12/2023*
Benedikt Rennekamp. Heidelberg University.	*10/2023*
Johannes Blumberg. Heidelberg University.	*10/2023*
Manuel Carrer. University of Oslo.	*09/2023*
Elham Kianiharchegani. Western University.	*08/2023*
Marc Stieffenhofer. Max Planck Institute for Polymer Research.	*06/2022*
Marius Bause. University of Amsterdam.	*01/2021*
Kiran H. Kanekal. Max Planck Institute for Polymer Research.	*12/2020*
Yasemin Bozkurt Varolgüneş. Koç University.	*05/2020*
Chan Liu. Max Planck Institute for Polymer Research.	*10/2019*

Journal referee *ACS Central Science*, *ACS Macro Letters*, *ACS Omega*, *Accounts of Chemical Research*, *Advanced Theory and Simulations*, *Advances in Physics: X*, *Biochimica et Biophysica Acta (BBA) - Biomembranes*, *Biointerphases*, *Biophysical Journal*, *Chemical Physics*, *Chemical Science*, *ChemistryOpen*, *Computer Physics Communications*, *EPL (Europhysics Letters)*, *Interface Focus*, *Journal of Applied Physics*, *Journal of Chemical Information and Modeling*, *Journal of Chemical Theory and Computation*, *Journal of Computational Chemistry*, *Journal of Computational Physics*, *Journal of Computational Science*, *Journal of Membrane Biology*, *Journal of Molecular Modeling*, *Journal of Physical Chemistry Letters*, *Langmuir*, *Machine Learning: Science and Technology*, *Molecular Systems Design & Engineering*, *Nature Communications*, *New Journal of Physics*, *Physical Chemistry Chemical Physics*, *Physical Review Letters*, *Plos One*, *Polymer Crystallization*, *SciPost Physics*, *Science Advances*, *Scientific Reports*, *Soft Matter*, *The Journal of Chemical Physics*, *The Journal of Physical Chemistry*

Publications

84. Magdalena Szczuka, Gilberto P. Pereira, Luis J. Walter, Marc Gueroult, Pierre Poulain, Tristan Bereau, Paulo C. T. Souza, Matthieu Chavent. Fast Parametrization of Martini3 Models for Fragments and Small Molecules. *J. Chem. Theory Comput.* **22** (2025)

83. Daniel Nagel, Tristan Bereau. Fokker–Planck Score Learning: Efficient Free-Energy Estimation under Periodic Boundary Conditions. *J. Phys. Chem. B* **129** (2025)

82. Luis J. Walter, Tristan Bereau. Navigating chemical space: multi-level Bayesian optimization with hierarchical coarse-graining. *Chem. Sci.* **16** (2025)

81. Bálint Máté, François Fleuret, Tristan Bereau. Solvation free energies from neural thermodynamic integration. *The Journal of Chemical Physics* **162** (2025)

80. Tristan Bereau, Luis J. Walter, Joseph Rudzinski. Martignac: Computational Workflows for Reproducible, Traceable, and Composable Coarse-Grained Martini Simulations. *J. Chem. Inf. Model.* **64** (2024)

Streamlit app

Github repository

Online talk

79. Bálint Máté, François Fleuret, Tristan Bereau. Neural Thermodynamic Integration: Free Energies from Energy-Based Diffusion Models. *J. Phys. Chem. Lett.* **15** (2024)

    Github repository

    Online talk

78. Andrea Anelli, Hanno Dietrich, Philipp Ectors, Frank Stowasser, Tristan Bereau, Marcus Neumann, Joost Van Den Ende. Robust and efficient reranking in crystal structure prediction: a data driven method for real-life molecules. *CrystEngComm* **26** (2024)

77. Ganna Gryn'Ova, Tristan Bereau, Carolin Müller, Pascal Friederich, Rebecca C. Wade, Ariane Nunes-Alves, Thereza A. Soares, Kenneth M. Merz, Jr.. EDITORIAL: Chemical Compound Space Exploration by Multiscale High-Throughput Screening and Machine Learning. *J. Chem. Inf. Model.* **64** (2024)

76. Stefan Bauer, Peter Benner, Tristan Bereau, Volker Blum, Mario Boley, Christian Carbogno, C Richard A Catlow, Gerhard Dehm, Sebastian Eibl, Ralph Ernstorfer, Ádám Fekete, Lucas Foppa, Peter Fratzl, Christoph Freysoldt, Baptiste Gault, Luca M Ghiringhelli, Sajal K Giri, Anton Gladyshev, Pawan Goyal, Jason Hattrick-Simpers, Lara Kabalan, Petr Karpov, Mohammad S Khorrami, Christoph T. Koch, Sebastian Kokott, Thomas Kosch, Igor Kowalec, Kurt Kremer, Andreas Leitherer, Yue Li, Christian Liebscher, Andrew Logsdail, Zhongwei Lu, Felix Luong, Andreas Marek, Florian Merz, Jaber R. Mianroodi, Jörg Neugebauer, Zongrui Pei, Thomas A R Purcell, Dierk Raabe, Markus Rampp, Mariana Rossi, Jan-Michael Rost, James Saal, Ulf Saalmann, Kasturi Narasimha Sasidhar, Alaukik Saxena, Luigi Sbailò, Markus Scheidgen, Marcel Schloz, Daniel F Schmidt, Simon Raphael Teshuva, Annette Trunschke, Ye Wei, Gerhard Weikum, Rui Patrick Xian, Yi Yao, Junqi Yin, Meng Zhao, Matthias Scheffler. Roadmap on data-centric materials science. *Modelling Simul. Mater. Sci. Eng.* **32** (2024)

    Featured in ChemEurope.com

75. Luca M Ghiringhelli, Carsten Baldauf, Tristan Bereau, Sandor Brockhauser, Christian Carbogno, Javad Chamanara, Stefano Cozzini, Stefano Curtarolo, Claudia Draxl, Shyam Dwaraknath, Ádám Fekete, James Kermode, Christoph T. Koch, Markus Kühbach, Alvin Noe Ladines, Patrick Lambrix, Maja-Olivia Himmer, Sergey V. Levchenko, Micael Oliveira, Adam Michalchuk, Ronald E. Miller, Berk Onat, Pasquale Pavone, Giovanni Pizzi, Benjamin Regler, Gian-Marco Rignanese, Jörg Schaarschmidt, Markus Scheidgen, Astrid Schneidewind, Tatyana Sheveleva, Chuanxun Su, Denis Usvyat, Omar Valsson, Christof Wöll, Matthias Scheffler. Shared metadata for data-centric materials science. *Sci Data* **10** (2023)

74. Kübra Kaygisiz, Lena Rauch-Wirth, Arghya Dutta, Xiaoqing Yu, Yuki Nagata, Tristan Bereau, Jan L. Münch, Christopher Synatschke, Tanja Weil. Data-mining unveils structure–property–activity correlation of viral infectivity enhancing self-assembling peptides. *Nat Commun* **14** (2023)

    Press release on Phys.org

    Press release on (idw) Informationsdienst Wissenschaft

73. Bernadette Mohr, Diego Van Der Mast, Tristan Bereau. Condensed-Phase Molecular Representation to Link Structure and Thermodynamics in Molecular Dynamics. *J. Chem. Theory Comput.* **19** (2023)

    Github repository

72. Kübra Kaygisiz, Arghya Dutta, Lena Rauch-Wirth, Christopher Synatschke, Jan L. Münch, Tristan Bereau, Tanja Weil. Inverse design of viral infectivity-enhancing peptide fibrils from continuous protein-vector embeddings. *Biomater. Sci.* **11** (2023)

    Gitlab repository

71. Martin Girard, Tristan Bereau. Induced asymmetries in membranes. *Biophysical Journal* **122** (2022)

70. Arghya Dutta, Tristan Bereau, Thomas A. Vilgis. Identifying Sequential Residue Patterns in Bitter and Umami Peptides. *ACS Food Sci. Technol.* **2** (2022)

    Github repository

69. Marc Stieffenhofer, Christoph Scherer, Falk May, Tristan Bereau, Denis Andrienko. Benchmarking coarse-grained models of organic semiconductors via deep backmapping. *Front. Chem.* **10** (2022)

68. Kiran Kanekal, Joseph Rudzinski, Tristan Bereau. Broad chemical transferability in structure-based coarse-graining. *The Journal of Chemical Physics* **157** (2022)

67. Isabel Kleinwächter, Bernadette Mohr, Aljoscha Joppe, Nadja Hellmann, Tristan Bereau, Heinz D. Osiewacz, Dirk Schneider. CLiB – a novel cardiolipin-binder isolated *via* data-driven and *in vitro* screening. *RSC Chem. Biol.* **3** (2022)

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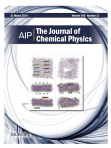

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